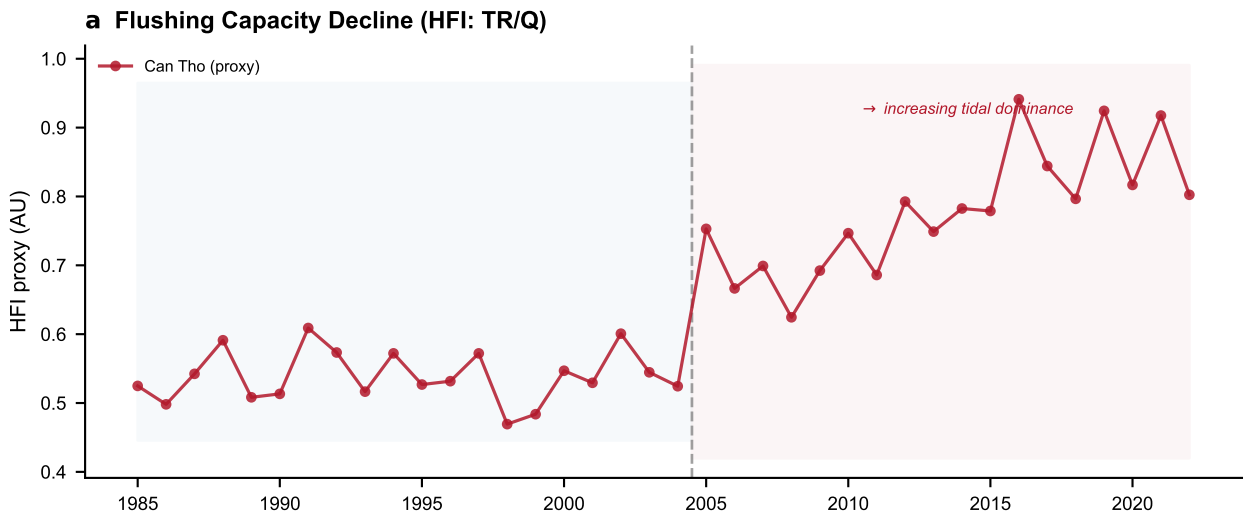
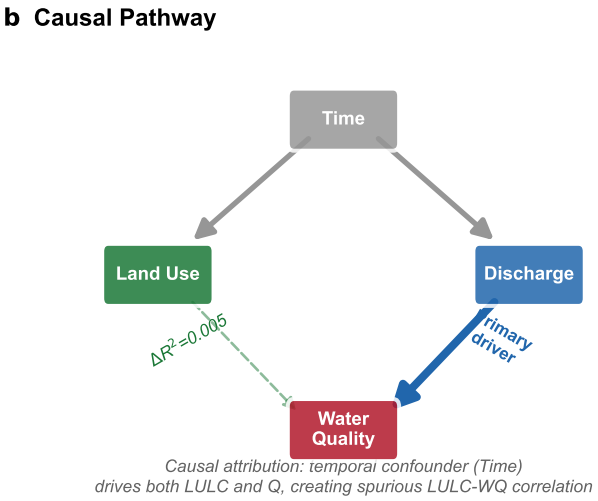
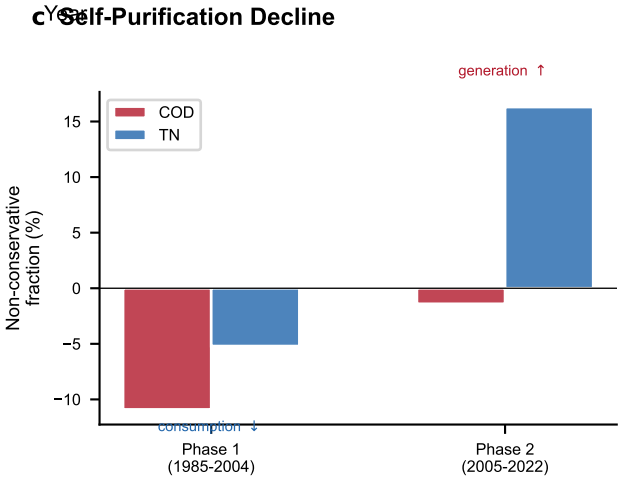



Causal attribution: temporal confounder (Time) drives both LULC and Q, creating spurious LULC-WQ correlation




## d Policy Implications

### Retention-Aware Delta Management

**Current approach: Land-use regulation**

- Restrict aquaculture expansion
- Assumption: LULC → WQ (confounded by time)
- $\Delta R^2 = 0.005 \Rightarrow$ **98% temporal confounding**

*Evidence-based redirection*

**Proposed: Hydrodynamic flushing management**

1. Upstream reservoir flushing pulses during dry season
2. Sediment bypass at dams to restore natural transport
3. Tidal barrier management to optimize residence time ($\tau$)
4. Real-time Q-normalized WQ monitoring (not just concentration)

⇒ ***Target the conveyor belt (hydrology), not just the cargo (land use)***